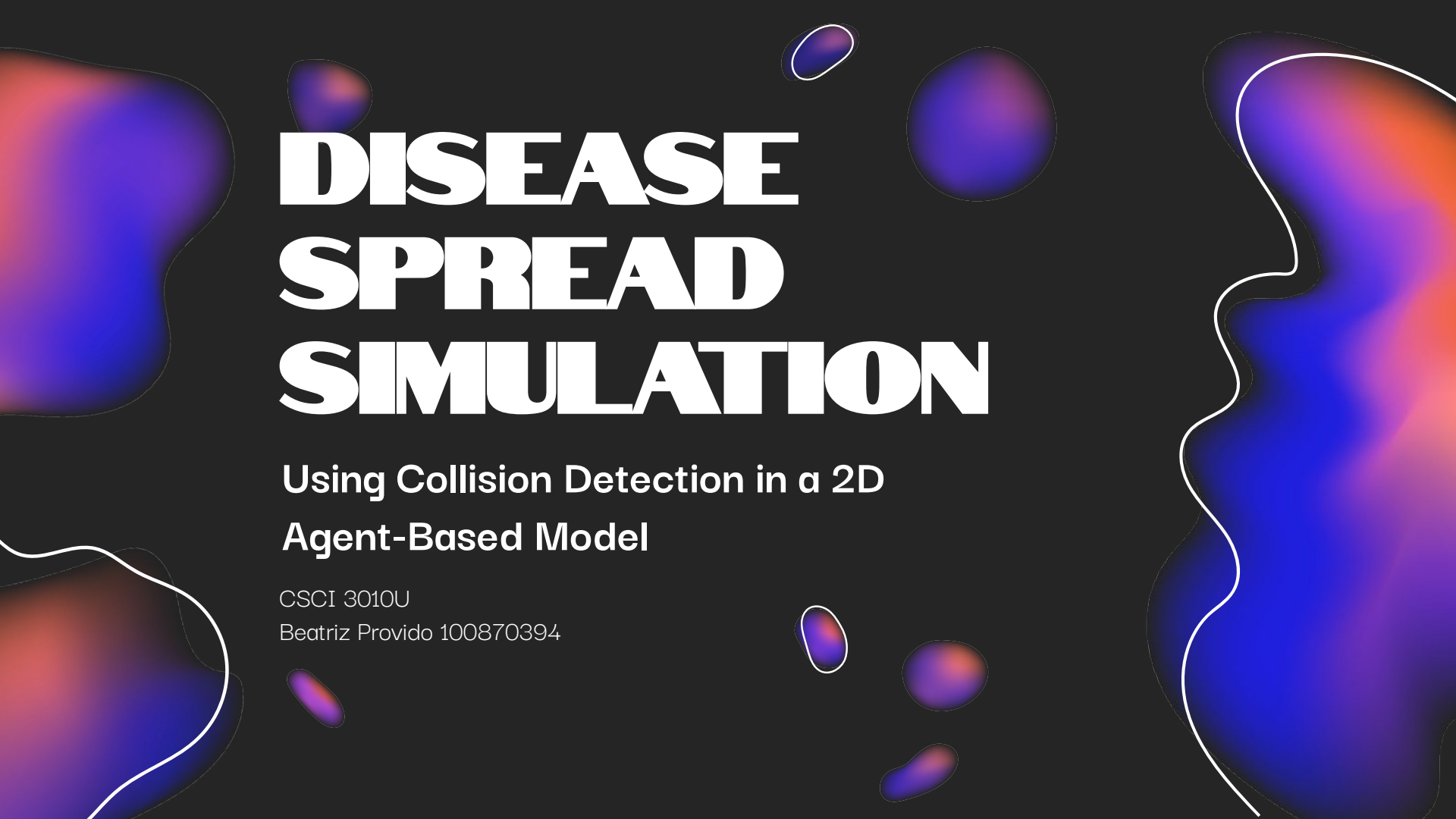


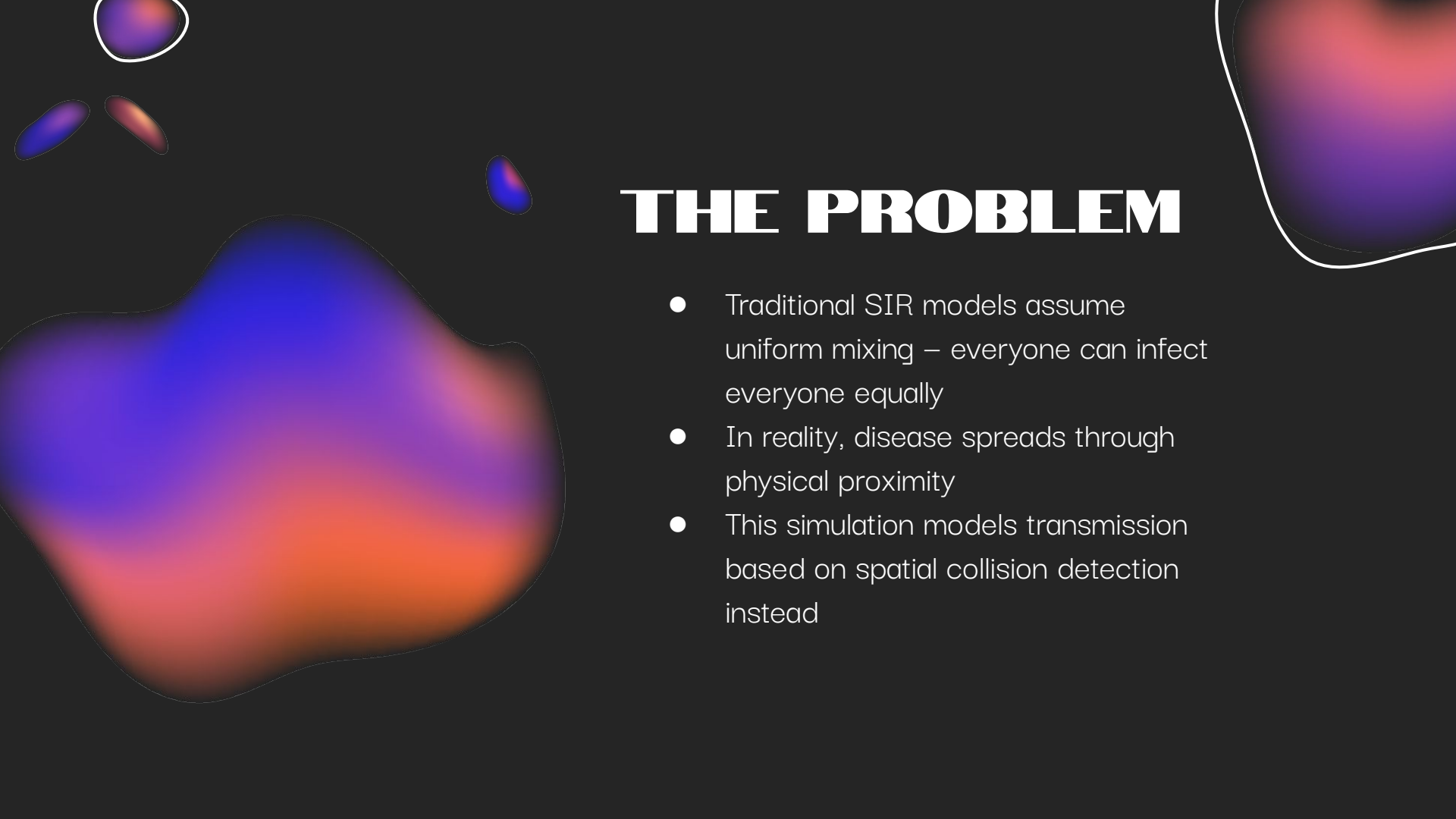
The background is a dark charcoal gray. It features several abstract, organic shapes in shades of blue, purple, and red. Some of these shapes have white outlines, giving them a 3D or cut-out appearance. The shapes are scattered across the slide, with larger ones on the left and right sides, and smaller ones near the text.

DISEASE SPREAD SIMULATION

**Using Collision Detection in a 2D
Agent-Based Model**

CSCI 3010U

Beatriz Provideo 100870394

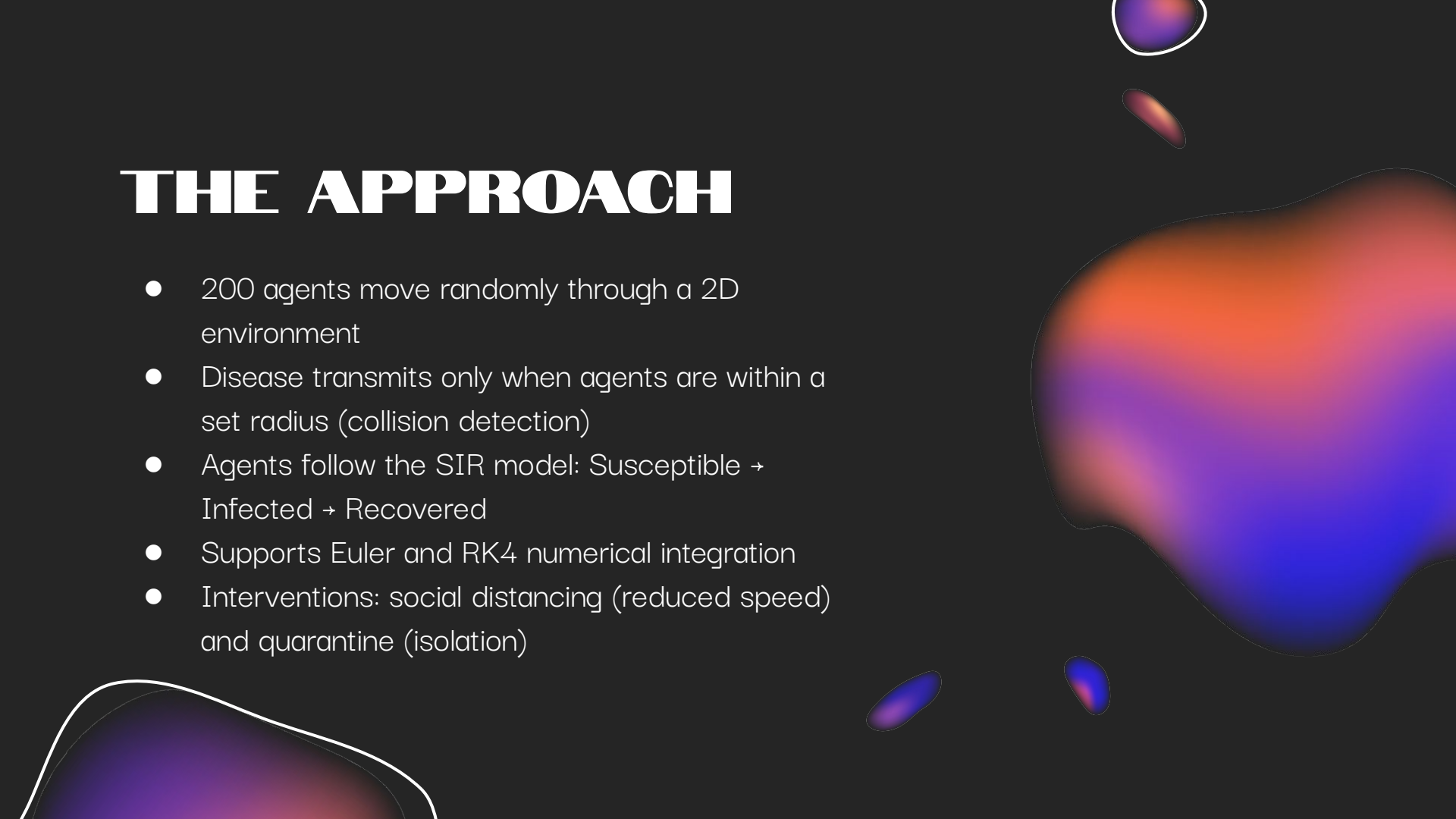
The background is dark grey with several abstract, colorful blobs. A large, irregular blob in the lower-left transitions from blue at the top to orange at the bottom. A smaller, similar blob is in the upper-left. A large, rounded blob in the upper-right transitions from purple to pink. A small, elongated blue and orange blob is in the middle-left. A small, circular blue and orange blob is in the top-left. A small, elongated blue and orange blob is in the middle-right.

THE PROBLEM

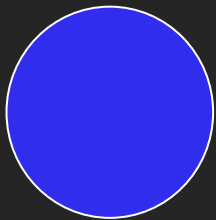
- Traditional SIR models assume uniform mixing – everyone can infect everyone equally
- In reality, disease spreads through physical proximity
- This simulation models transmission based on spatial collision detection instead

THE APPROACH

- 200 agents move randomly through a 2D environment
- Disease transmits only when agents are within a set radius (collision detection)
- Agents follow the SIR model: Susceptible → Infected → Recovered
- Supports Euler and RK4 numerical integration
- Interventions: social distancing (reduced speed) and quarantine (isolation)

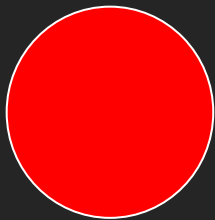


HOW THE SIMULATION WORKS



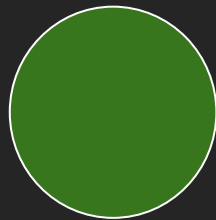
SUSCEPTIBLE

Healthy, vulnerable
to disease



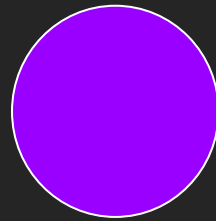
INFECTED

Sick, spreads
disease to nearby
agents



RECOVERED

Immune, healed from
disease + cannot get
sick again

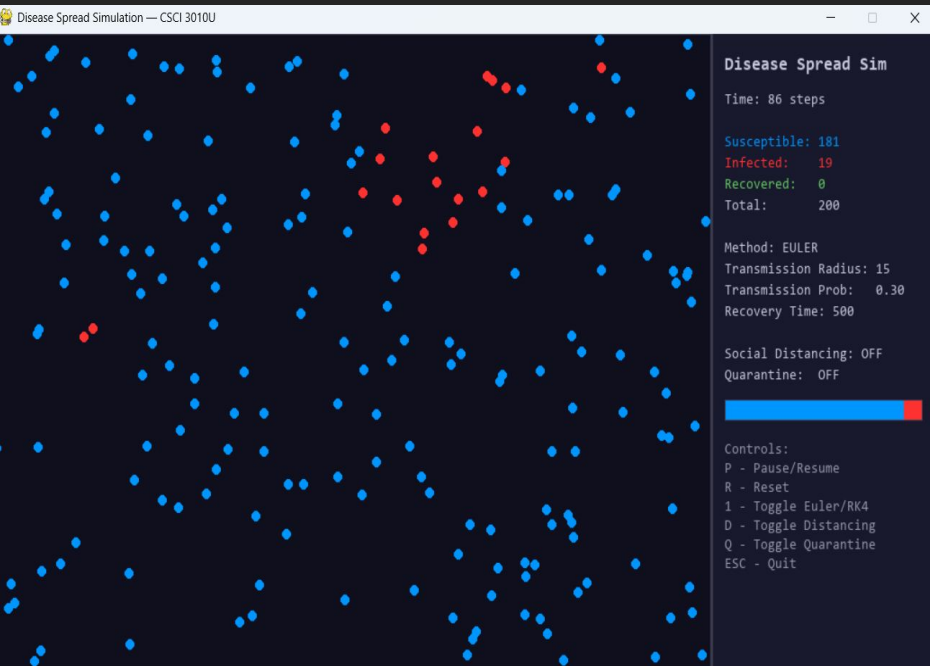


QUARANTINED

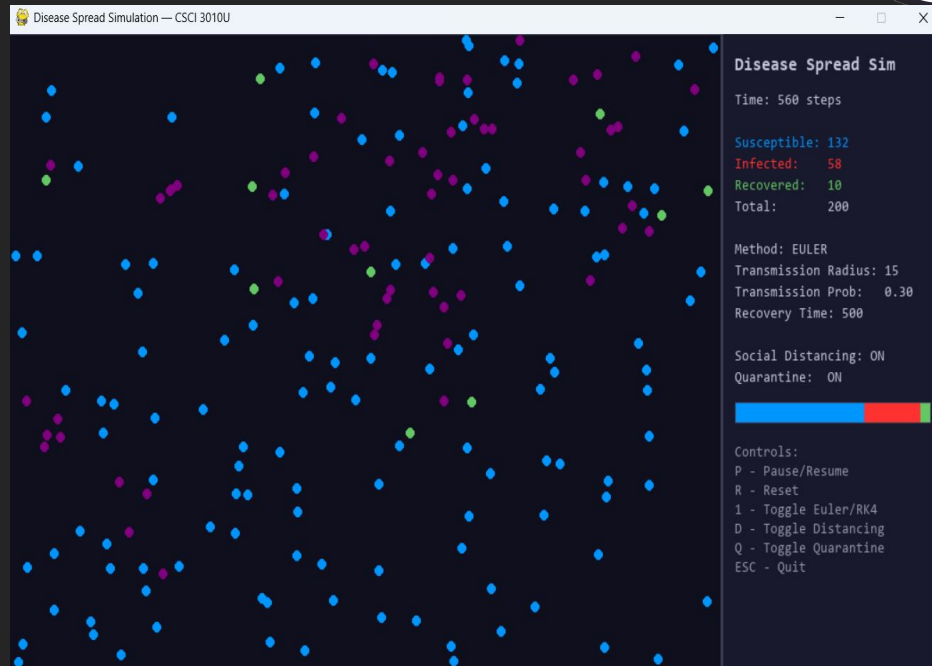
Isolated, stops
moving

- Agents move randomly in 2D, bouncing off walls
- Infection occurs within 15px radius at 30% probability
- Recovery after 500 time steps

SIMULATION IN ACTION

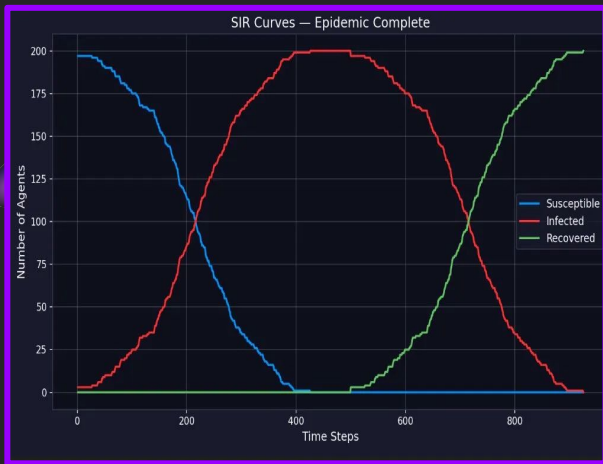


Baseline



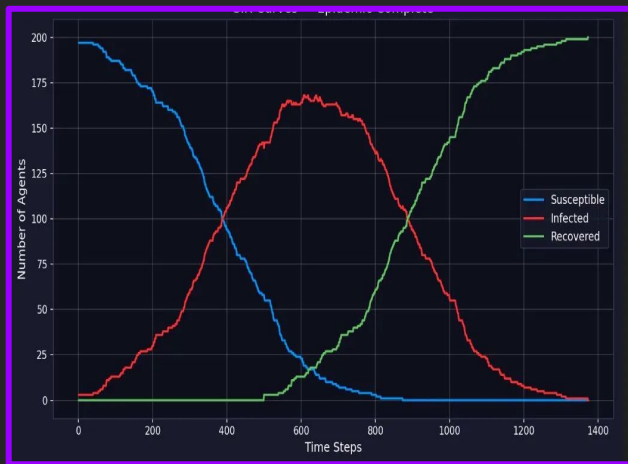
Social Distancing + Quarantine

EXPERIMENT RESULTS



Baseline:

- Peak at ~200 infected
- Ends at ~900 steps



Social Distancing ON:

- Peak at ~170 infected
- Ends at ~1400 steps



Quarantine

- Disease barely spreads

EULER VS RK4 INTEGRATION

EULER

- Simpler, faster method
- Updates position in a straight line each step
- Works well for constant velocity

Note: With zero forces, both methods produce identical results. Minor differences in quarantine runs come from randomness, not the integration method

RK4

- Samples velocity at 4 points per step
- More accurate for curved trajectories
- Advantage appears when forces are present

CONCLUSION & FUTURE WORK

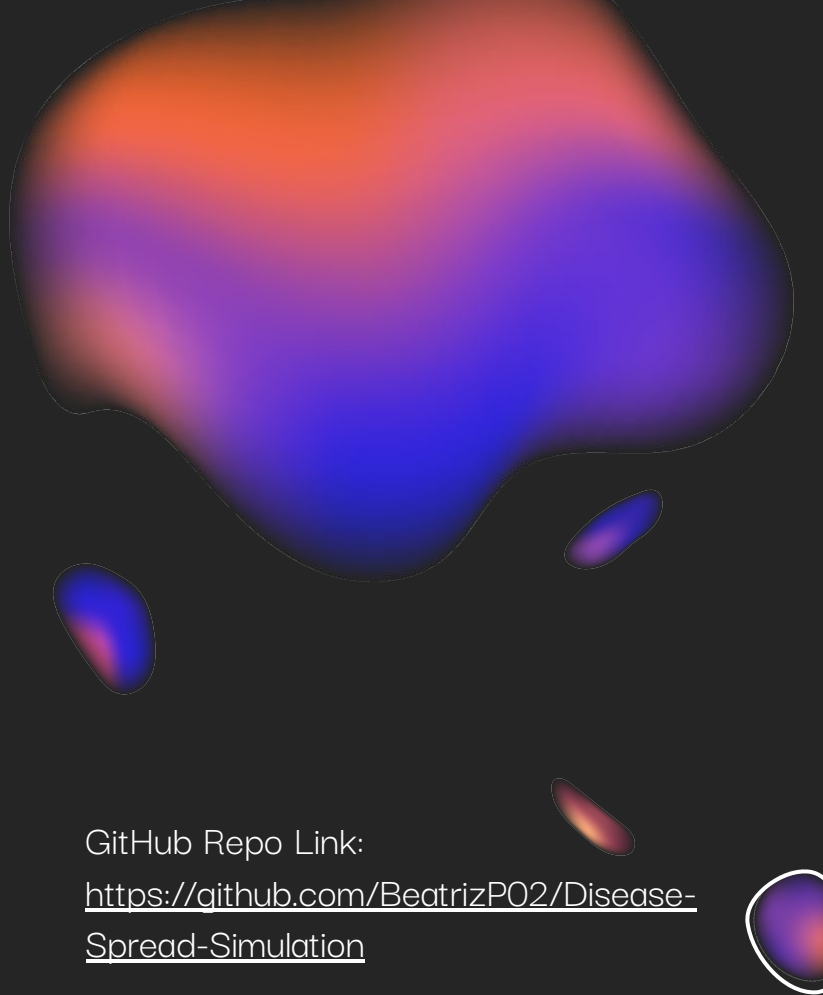
- Spatial proximity-based transmission produces realistic SIR dynamics
- Social distancing flattens the curve, quarantine suppresses spread
- Euler and RK4 produce identical results under zero-force conditions

Future Work:

- Add repulsion forces to differentiate Euler vs RK4
- Implement spatial partitioning for better performance scaling
- Compare against traditional ODE-based SIR model

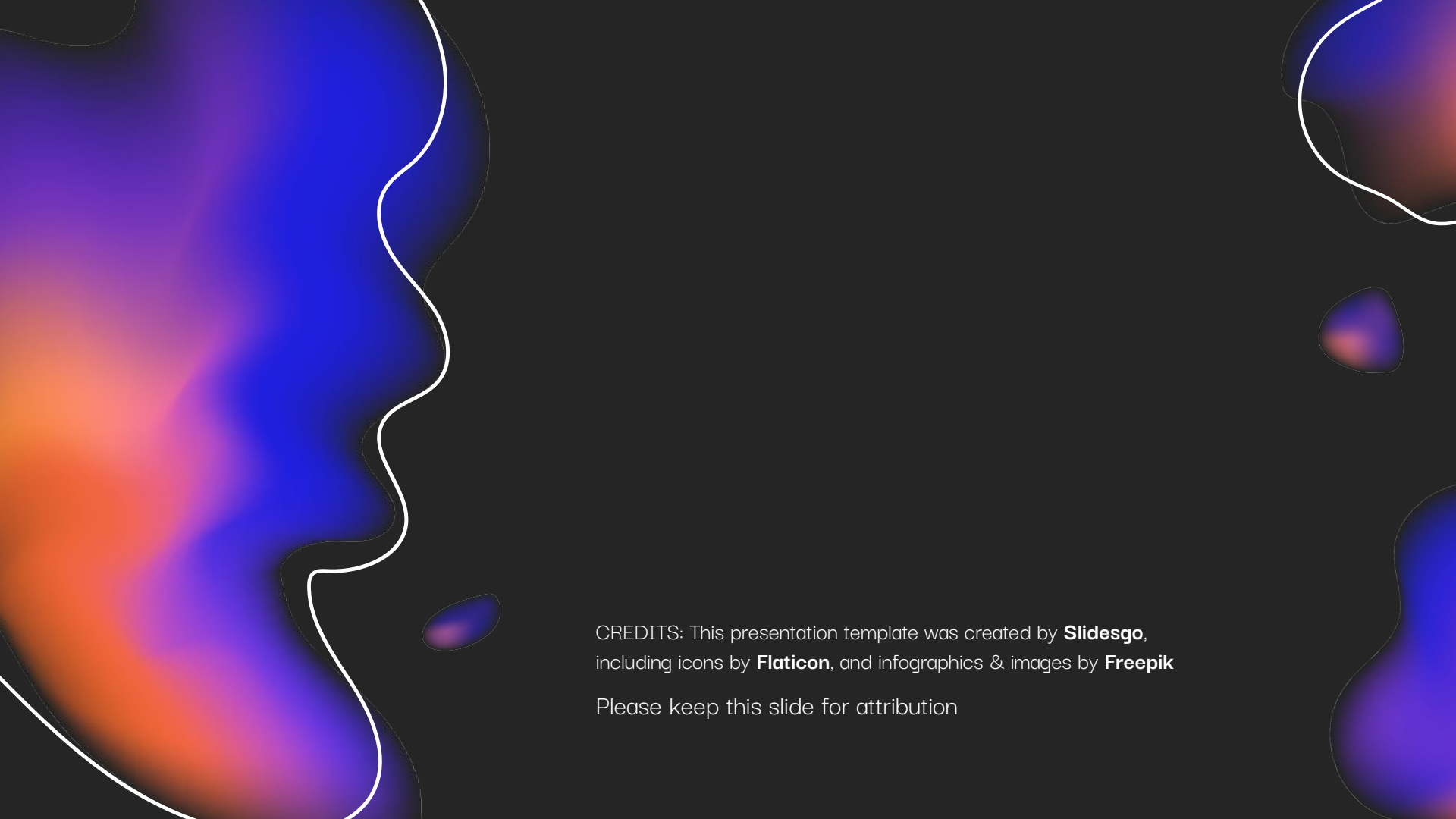
GitHub Repo Link:

<https://github.com/BeatrizP02/Disease-Spread-Simulation>





**THANK
YOU FOR
LISTENING**



CREDITS: This presentation template was created by **Slidesgo**,
including icons by **Flaticon**, and infographics & images by **Freepik**

Please keep this slide for attribution